

incorporate a multiphase nanocomposite oxygen■scavenging additive, analogous to the MNCP, into the pouch■cell electrolyte or separator to actively oxidize residual oxygen during formation cycling, thereby preventing gas evolution

Executive Technology Brief

Domain: energy_harvesting

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'gas-induced swelling' concerning lithium-ion pouch cell. User-submitted engineering problem (session input): "Why do lithium-ion pouch cells blister and swell during formation cycling before any external abuse, and how can we suppress gas-induced swelling without adding a vent?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Understanding the thermal stability of high-energy density alloy anodes is critical for the safe deployment of sodium-ion batteries (SIBs). Here, accelerating rate calorimetry (ARC), post-mortem chara'; 'Formation of robust solid/cathode electrolyte interphases (S/CEI) is vital for long-term stability and high-performance operation of lithium-ion batteries (LIBs), particularly under high voltage regim'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: oxygen oxidation (as a chemical reaction that consumes oxygen and does not generate gas)

Architecture: PROPOSED architecture (no physical system exists): ambient source characterization element; transduction stage (piezo / thermo / inductive); power conditioning and storage buffer; load interface + duty-cycle governor — realizing: incorporate a multiphase nanocomposite oxygen■scavenging additive, analogous to the MNCP, into the pouch■cell electrolyte or separator to actively oxidize residual oxygen during formation cycling, thereby preventing gas evolution

Governing Model: Energy harvesting / power conversion governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

ENH-001: Seebeck effect (thermoelectric voltage). Expression: $V = S_{diff} * dT$. Applicability to this invention: CONDITIONAL; holds under: couple materials characterized; maintained temperature difference across the junctions. Assumptions: S_{diff} is material- and temperature-dependent — source the value for the actual couple; output scales with the MAINTAINED difference, not with ambient temperature alone. Engaged variables: NOT ESTABLISHED.

ENH-002: Piezoelectric constitutive relation (charge form). Expression: $D = d_T * T_{stress} + \epsilon_{ps_T} * E$. Applicability to this invention: CONDITIONAL; holds under: small-signal linear regime below depoling limits. Assumptions: coupling coefficients are frequency-, load- and temperature-dependent; quoted values are not design values until measured in situ. Engaged variables: NOT ESTABLISHED.

ENH-003: Maximum power transfer to a load. Expression: $P_{max} = V_{oc}^2 / (4 * R_{int})$. Applicability to this invention: CONDITIONAL; holds under: linear Thevenin-equivalent source; matched load. Assumptions: real conditioning chains never achieve the theoretical maximum — measure the achieved fraction. Engaged

variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.